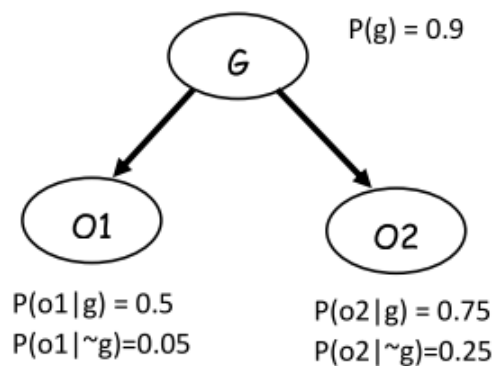


Bayes Theorem Pop Quiz User Guide

Requirements:

- 2 .txt files:
 - Probabilities
 - Neighbors



Probabilities File Format:

example_prob.txt

```
g:0.9
o1|g:0.5
o1|~g:0.05
o2|g:0.75
o2|~g:0.25
```

- Each node should be paired with a colon and a probability without spaces.

- It is not required to upload all possible probabilities, the algorithm is capable of calculating them.
- [NODE]:[PROBABILITY]
- [NODE A][NODE B]:[PROBABILITY]
- [NODE A][NODE B][NODE C]:[PROBABILITY]
- [NODE A][NODE B][NODE C]:[PROBABILITY]

Neighbors File Format:

example_neighbors.txt

g:o1,o2

- The node before the colon is the parent.
- The node(s) after the colon are the children/neighbors.
- Each parent node should be paired with a colon and a list of all its children.
- [NODE]:[LIST OF NODE CHILDREN]

Functions:

```
def parse_input(prob_file,neh_file):
```

- This function takes the names of two files as arguments. It parses the inputs with file handling just in case the file does not work. All the probabilities are converted to floats. The data structure returned by this method is a list of two dictionaries. The dictionary breakdown: list[0] = probabilities, list[1] = neighbors → [{string : float},{string : string}]

```
def all_possibilities(grf):
```

- This function generates all the complements of the data given in the original text file, and puts the added probabilities into the original graph. It takes the graph created from the parse_input() method. This simplifies some of the calculations.

```
def calculate_probability(user_input,grf):
```

- This function calculates the probability, given the user input and graph data structure from before. It returns a float. It iterates through the grf parameter to determine if the given input is already in the data structure. If not, it uses methods such as marginalization and normalization to find the probability of the given input.

```
def joint_probability(event,prob):
```

- This function calculates the joint probability given the event (user input split by the “|” symbol) and the probabilities dictionary. It multiplies the probability by what is found in the dictionary.

```
def main():
```

- This function is where the user input occurs. It asks about file names and probabilities. It has a built-in verification to make sure that the file and input is valid. It prints the output cleanly for users.

```
def cond_prob(arg,graph):
```

- This function is where the bulk of the calculations occur. It finds the numerator using the joint probability method as well as calculates the denominator, and then returns it.

How to Use:

1. Input the name of the **probabilities** file when the program prompts you. Do not include the “.txt” at the end. (**Welcome to the Bayes Network Builder. What is the name of the file with your PROBABILITIES? See the user guide for the specific format inside the file.**)
2. Input the name of the **neighbors** file when the program prompts you to. Do not include the “.txt” at the end. (**What is the name of the file with the NODES**

AND THEIR CHILDREN? See the user guide for the specific format inside the file.)

3. The program will display the graph as a list with two dictionaries. The first dictionary contains the probabilities, and the second dictionary contains the neighbors. Type “y” to verify the data, and “n” to reupload the files if they are incorrect.

4. Input the desired probabilities you want to calculate.
(Calculate probability. Type your desired probability as specified in the user guide.)

The format is as follows:

- a. **A / A|B / A,B / A|B,C / A,B|C**
 - b. If you are looking for $P(A)$ for example, only input A.
 - c. Only input what is in the parentheses with no spaces.
5. Now you have the probability. You can do it as many times as you want. Type stop to stop.

```
PS G:\My Drive\AI\Unit 5> python Yeboah_Frances_U5_L1.py
Welcome to the Bayes Network Builder. What is the name of the file with your PROBABILITIES? See the user guide for the specific format inside the file.
pop_quiz_prob
What is the name of the file with the NODES AND THEIR CHILDREN? See the user guide for the specific format inside the file.
pop_quiz_nehs
This is your graph of probabilities. [{'g': 0.9, 'o1|g': 0.5, 'o1|~g': 0.05, 'o2|g': 0.75, 'o2|~g': 0.25, '~g': 0.09999999999999998, '~o1|g': 0.5, '~o1|~g': 0.95, '~o2|g': 0.25, '~o2|~g': 0.75}, {'g': 'o1,o2'}]
If this is correct, type 'y.' Else, type 'n.'
y
Calculate probability. Type your desired probability as specified in the user guide.
~o1|g
P( ~o1|g ) = 0.5
```